Check for updates

ACM DIGITAL LIBRARY

Association for Computing Machinery

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3700297.3700385

RESEARCH-ARTICLE

Research on Student Performance Prediction Based on Random Forest Algorithm

SIHONG PAN, Xiamen University of Technology, Xiamen, Fujian, China

WEI DAI, Sichuan University, Chengdu, Sichuan, China

Open Access Support provided by:

Xiamen University of Technology

Sichuan University

PDF Download
3700297.3700385.pdf
01 March 2026
Total Citations: 0
Total Downloads: 397

Published: 18 November 2024

Citation in BibTeX format

ISAIE 2024: 2024 International Symposium on Artificial Intelligence for Education
September 6 - 8, 2024
Xi'an, China

Research on Student Performance Prediction Based on Random Forest Algorithm

Sihong Pan
Academic Affairs Office, Xiamen University of Technology
China
2573414579@qq.com

Wei Dai*
Sichuan University Jinjiang College
China
2675000730@qq.com

Abstract

With the increasing attention to the analysis and prediction of student achievement in the field of education, educational data mining (EDM) has become the focus of research. This paper proposes a student performance prediction model based on random forest algorithm, which aims to improve the accuracy and interpretability of the prediction model by integrating multi-dimensional data features and deeply analyzing the impact of different feature data on the prediction results. The study data included students' campus card data, library data and student status data, and the study subjects were 105 students majoring in civil engineering at a university. Experimental results show that the prediction accuracy of the model on the training set and the test set is 93.15% and 77.42%, respectively. The study found that the important characteristics that affected the prediction results included the average monthly borrowing volume, late return status, average monthly breakfast frequency, weighted average grade in the previous semester, and weighted average grade of professional courses. The research in this paper provides scientific data support and decision-making basis for education managers, which is helpful to optimize and improve the quality of teaching.

CCS Concepts

• **Applied computing** → Operations research; Forecasting.

Keywords

EDM, Random Forest Algorithm, Student Achievement Prediction Model, Multidimensional data features

ACM Reference Format:
Sihong Pan and Wei Dai. 2024. Research on Student Performance Prediction Based on Random Forest Algorithm. In *2024 International Symposium on Artificial Intelligence for Education (ISAIE 2024), September 06–08, 2024, Xi'an, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3700297.3700385

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ISAIE 2024, September 06–08, 2024, Xi'an, China
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0710-0/24/09
https://doi.org/10.1145/3700297.3700385

1 Introduction

With the increasing attention to education from all walks of life, scientific analysis of students' current performance and accurate prediction have become the focus of education researchers. Educational Data Mining (EDM) refers to the use of evolving technologies and methods to explore data in a specific educational environment, with the aim of predicting student achievement through data mining, so as to provide effective decision support for students and instructional administrators. The traditional teaching mode is generally to conduct post-event analysis and early warning for unqualified students after the results are announced, which has a certain lag and cannot meet the requirements of today's high-quality talent training. With the gradual promotion of digital campus construction, the school has obtained a large amount of student information, and on the basis of protecting students' privacy, it uses artificial intelligence technology to predict students' academic performance and carry out early warning of results, which is helpful to optimize and improve the quality of teaching. In recent years, with the development of artificial intelligence technology, machine learning has been widely used in the research direction of student performance prediction, through data collection and preprocessing, feature engineering, model selection and training, model evaluation and optimization and other steps, it can provide personalized learning support for students, provide teachers with data-driven teaching improvement suggestions, and ultimately improve teaching quality and student learning effect. [1]

2 Related Research

In recent years, with the development of artificial intelligence technology, machine learning has been widely used in the research direction of student achievement prediction. Zhang Feng et al. [2] introduced the basic process of constructing a student performance prediction model based on machine learning technology, and systematically sorted out two types of learning data features: non-process data features and process data features. Liu Lijuan et al. [3] constructed a prediction model of students' academic early warning based on the Random Forest (RF) algorithm, and quantified and compared the influence and importance of multi-dimensional features on academic early warning prediction. Li Kaiwei [4] proposed a method to predict student performance by using student behavior data, and the model optimization was carried out through Stacking ensemble, which has certain research significance for auxiliary teaching management. Alsariera et al. [5] classified the attributes used to evaluate academic performance into seven categories: demographic factors, academic, internal assessment, communication, behavior, psychological, and family/personal attributes. Among these, the most commonly used attributes are

academic factors such as attendance, grade point average (GPA), and final grades [6]. These variables are the most significant and direct, and they hold substantial value for future educational and career mobility [7]. Another frequently used attribute is students' demographic factors (including gender, age, and nationality). In China, many researchers have applied student behavior factors, such as campus card data, to predict academic performance. Lan Jiafeng [8], based on campus card data, analyzed the various behaviors of freshmen to predict and warn about their academic status. Sun Meijuan et al. [9] conducted an associative analysis of campus card data and student grades to explore the relationship between student behavior and academic performance, creating "profiles" of students. Additionally, course factors, such as the sequence of courses and related course grades, also impact student performance. The foundational knowledge accumulated in previous courses is applied in subsequent courses, indicating that different courses are interconnected. This study aims to uncover these potential connections and apply them to performance prediction to enhance the accuracy of predictive models.

Regarding predictive algorithms, Liu Xiaoyun et al. [10] used multiple linear regression methods to predict the graduation grades of computer application majors based on their first-year grades, achieving high prediction accuracy. Lin Tingting [11] applied the backpropagation neural network algorithm to estimate the average college entrance examination scores of a 2016 high school cohort, with the maximum absolute error of the predictions being only 2 points. Naser et al. used an artificial neural network (ANN) model, incorporating factors such as college entrance exam scores, university GPA, and gender as indicators. Using a multilayer perceptron topology model for training data, they achieved an accuracy rate of over 80% in predicting student performance. Alsariera et al. statistically analyzed the most commonly used academic performance prediction machine learning algorithms from 2015 to 2022, including artificial neural networks, decision trees (DT), support vector machines (SVM), and linear regression (LinR). They compared the factors and prediction accuracy of various algorithms, finding that ANN and DT had the highest accuracy.

At present, machine learning mainly uses students' learning behaviors to make predictions, but students' information data dimensions are generally small, and the data features cannot be fully explored, which affects the accuracy of model prediction. In addition, there is a lack of interpretability of different feature data to the prediction results of the model. Based on this background, this paper proposes a student performance prediction model based on random forest algorithm. By integrating multi-dimensional data features and conducting in-depth analysis of the impact of different feature data on the prediction results of the model, the aim is to improve the accuracy and interpretability of the prediction model, and provide more scientific data support and decision-making basis for education managers.

3 Random forest algorithm

In the prediction model of data mining, classification and regression tasks are usually included, and random forest algorithms can effectively deal with these two types of problems. In this paper, student achievement prediction is treated as a multi-categorical problem. Compared with a single decision tree, the random forest algorithm can reduce the variance of a single decision tree to a certain extent, and improve the stability and generalization ability of the model. The steps of the random forest algorithm are as follows:

Self-service sampling: Random sampling with return from the original training dataset generates multiple self-service sample sets that will be used to build decision trees.

Build a decision tree: Use decision tree algorithms (such as CART, ID3, etc.) to build a decision tree for each self-service sample set. These decision trees are independent during training and do not affect each other.

Feature selection: When building a decision tree, each node is split using a randomly selected subset of features, rather than using all features. This method can reduce the overfitting phenomenon of the model and improve the generalization ability.

Ensemble Prediction: After all decision trees are built, new input samples are fed into each decision tree for prediction. Each tree will give a classification result, and the random forest algorithm will integrate these results through a voting mechanism to finally output the most likely classification result.

A performance prediction model was constructed based on random forest algorithm

In this study, 105 students in three classes of 2022 majoring in civil engineering at a university were studied, and the random forest algorithm was used to model the students' behavior data to predict the students' end-of-semester grades. The research data include all-in-one card data (consumption, access control), library data (borrowing, arrival) and academic affairs data (students, academic status). The data was collected from February to June 2022, that is, in the second semester of the 2021-2022 academic year.

3.1 Data preparation and preprocessing

The student data used in this study comes from the databases of relevant business systems within the university. The campus card system uses SQL Server, while the library, access control, and academic data are managed by Oracle. After preprocessing the business system data, the following 11 feature data points were selected:X1: Gender (0 for male, 1 for female)X2: Source of students (1 for regular high school students, 2 for vocational school students, 3 for integrated secondary and higher vocational students)X3: CET-4 passing status (0 for no, 1 for yes)X4: Average monthly book borrowing volume (numerical)X5: Average monthly library visits (numerical)X6: Late return status (numerical)X7: Average monthly spending (numerical)X8: Average monthly breakfast count (numerical)X9: Average monthly leave count (numerical)X10: Weighted average score of the previous semester (numerical)X11: Weighted average score of major courses (numerical) The target for prediction is the current semester's average grade of major courses (GRADE), classified as: G (80-100), P (60-80), and NP (below 60).

3.2 Random Forest Algorithm Model Implementation for Prediction

The programming language used for this experiment is R, running in the RStudio environment. After preprocessing, the dataset contains 105 samples and 11 variables. The students' exam grades (grade) are treated as the response variable and set as a factor variable.

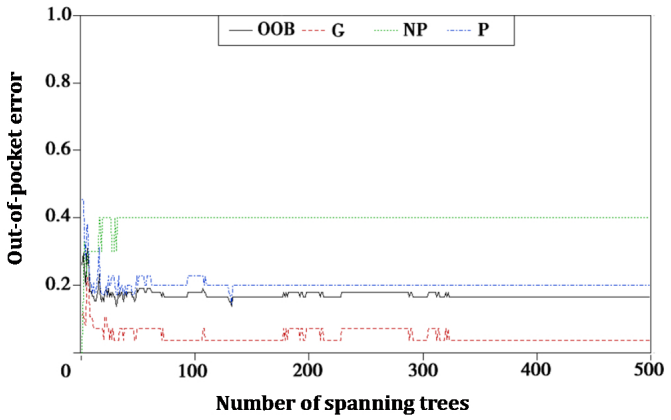

Figure 1: The relationship between the out-of-pocket error and the parameter ntree.

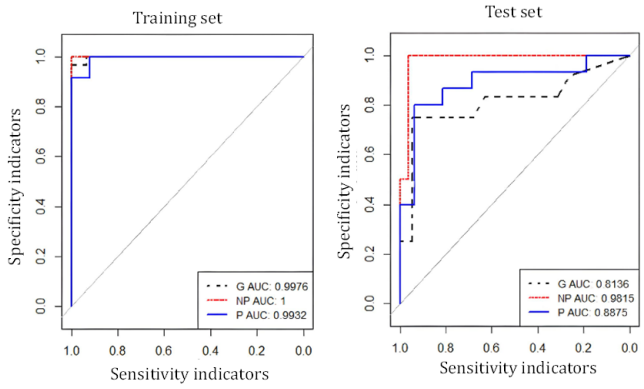

Figure 2: The training set is different from the test set, and the AUC is different.

In the experiment, 65% of the samples are selected as the training set, and the remaining as the test set. Using the randomForest function from the randomForest package, 76 samples are randomly drawn with replacement from the training set. At each node of each tree, 4 variables are randomly selected, generating a total of 200 decision trees. The mtry parameter in the model controls the number of features randomly chosen at each split. The default value, which is the square root of the total number of features, is typically recommended; thus, mtry is set to 4. The ntree setting affects the stability and accuracy of the model. Based on the relationship between ntree and error, when ntree is set to 200, the error value tends to stabilize (as shown in Figure 1).

The above steps have successfully constructed the grade prediction model cjyc_rf. Next, the predict function will be used with the cjyc_rf model and the test dataset (testdata) to generate predictions (testpredprob). Using testdata$GRADE, the confusionMatrix function will construct the confusion matrix and generate the ROC curve, which indicates the model's performance in grade prediction. Figures 2 and 3 show the predicted classification results for the training and test sets. Among the 73 training samples, 68 were correctly classified: G (excellent): 27, P (pass): 32, NP (fail): 9, with an accuracy of 0.9315. For the 33 test samples, 24 were correctly classified: G (excellent): 9, P (pass): 14, NP (fail): 1, with an accuracy of 0.7742. Although the cjyc_rf grade prediction model performs well on the test set, the small sample size and the random forest's tendency to overfit result in a noticeable discrepancy in accuracy between the training and test sets. The results as shown in Table 1.

By calling the importance function from the randomForest package and using the cjyc_rf model as a parameter, the importance of the 11 feature attributes in the prediction model was determined. The MeanDecreaseAccuracy and MeanDecreaseGini are two key metrics in the random forest model. They respectively represent the decrease in prediction accuracy and the impact of each variable on the heterogeneity of observations at each node of the classification trees. Higher values indicate greater importance of the corresponding feature.

As shown in Figure 3, the five features with the greatest impact on the prediction results are X5 (average monthly book borrowing volume), X6 (late return status), X8 (average monthly breakfast count), X10 (weighted average score of the previous semester), and X11 (weighted average score of major courses). The ranking of feature importance for grade prediction indicates that there is a certain continuity in the learning process, with the previous semester's performance significantly influencing subsequent learning stages. Additionally, the number of breakfast consumptions, late returns, and monthly library visits reflect the student's self-discipline and learning autonomy to some extent. However, factors such as the source of students, gender, and book borrowing frequency have relatively less impact compared to other dynamic data.

4 Conclusion

Prediction model by analyzing student behavior data and ranked the importance of these features. We selected the Random Forest algorithm, which is known for its efficiency in handling large datasets and high prediction accuracy. Additionally, it requires minimal preprocessing of data, such as standardization or normalization.

Table 1: Classification results for the training and prediction sets.

Training set confusion matrix				Prediction set confusion matrix			
PredictionReference	G	NP	P	PredictionReference	G	NP	P
G	27	0	3	G	9	0	1
NP	0	9	0	NP	0	1	0
P	1	1	32	P	3	3	14

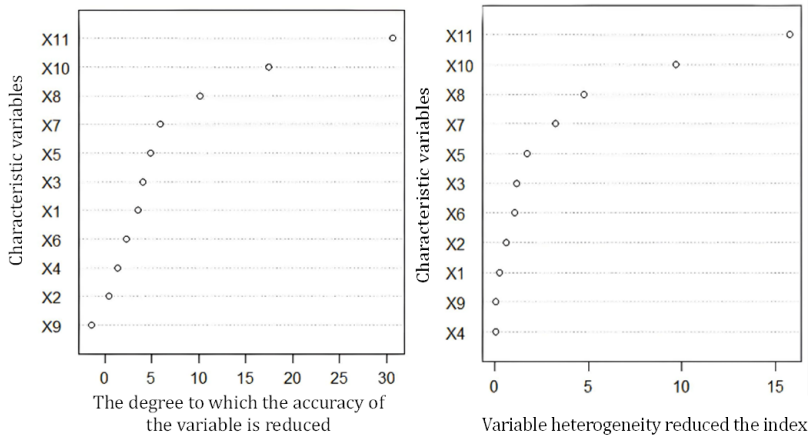

Figure 3: Ranking of the importance of feature variables.

However, the small sample size in this experiment led to some degree of overfitting. Despite this, the model provided valuable insights into the factors influencing student performance. Key features such as the average number of books borrowed per month, overdue book returns, the average number of breakfasts per month, and the weighted average grade of the previous semester had a significant impact on performance prediction. These findings highlight the continuity of the student's learning process, indicating that past academic performance strongly influences future outcomes. Moreover, behavioral indicators such as breakfast consumption, overdue book returns, and library visits reflect students' self-discipline and learning autonomy. In contrast, static characteristics such as student origin, gender, and borrowing frequency had a smaller impact on prediction results.

Based on these findings, we can draw several important conclusions:

1. Importance of Key Behavioral Features in Predicting Student Performance

The results of this study show that students' behavioral features, such as the average number of books borrowed per month, overdue book returns, and the average number of breakfasts per month, are significantly associated with their academic performance. These behavioral features reflect students' study habits and self-discipline, which in turn affect their academic performance. For example, students who borrow more books per month tend to engage in extracurricular reading and self-study, leading to better academic performance. Similarly, students with fewer overdue book returns usually have better time management skills, which also help them perform well academically.

2. Influence of Historical Academic Performance on Future Grades

The weighted average grade of the previous semester is another feature that significantly impacts prediction results. This finding suggests that students' academic performance has a certain continuity, with past grades playing a crucial role in predicting future performance. This result also supports the theory that "success breeds success," indicating that past successful experiences and achievements can boost students' confidence and motivation, leading to continued excellent performance in future studies.

3. Role of Student Self-Discipline and Learning Autonomy

The study found that students' self-discipline and learning autonomy have a significant impact on their academic performance. For example, students who eat breakfast more frequently each month usually pay more attention to regular living habits and physical health, which helps them maintain a good state for academic work. Additionally, students who visit the library more frequently usually exhibit more initiative in learning, which also contributes to their academic success. This finding emphasizes the importance of cultivating students' self-discipline and learning autonomy, and schools and parents should take measures to help students develop good study habits and living habits.

4. Application of Random Forest Algorithm in Educational Data Mining

This study demonstrates that the Random Forest algorithm is an effective tool for educational data mining, capable of accurately predicting student academic performance by effectively utilizing and ranking various behavioral features. Despite the challenges posed by the small sample size, the research still showcases the potential of machine learning technologies in enhancing educational assessment and intervention strategies. For instance, by predicting student performance, schools can identify students struggling academically early and provide targeted tutoring and support, thereby improving overall educational quality.

5. Limitations of the Study and Future Directions

Despite the achievements of this study, there are some limitations. First, the small sample size may lead to insufficient generalizability of the model, so future research should expand the sample size to improve the model's stability and accuracy. Second, this study only considered a limited number of features, and future research can introduce more features such as students' psychological state, family background, etc., to comprehensively understand the factors affecting student performance. Additionally, with the development of educational technology, future research can explore the application of more advanced machine learning algorithms such as

deep learning in student performance prediction to further enhance prediction accuracy and reliability.

In summary, this study conducted an in-depth analysis of student behavior data using the Random Forest algorithm, revealing key factors influencing student academic performance and demonstrating the potential application of machine learning technologies in educational data mining. Future research should further expand the dataset and feature range and explore more machine learning algorithms to continuously improve the performance of student performance prediction models, providing more scientific data support for educational decision-making.

References

[1] ZHANG Li, LU Xingning, LU Conglin, *et al.* Support vector machine in college entrance examination results pre. Journal of University of Science and Technology of China, 2017, 47(1):1-9

[2] Huy V C, Thu A V, Phuc H N. Powerlifting total score prediction based on an improved random forest regression algorithm[J]. Journal of Intelligent & Fuzzy Systems, 2024, 46(4):9999-10004.

[3] N. K, R. M, A. K V. XAI-Based Student Performance Prediction: Peeling Back the Layers of LSTM and Random Forest's Black Boxes[J]. SN Computer Science, 2023, 4(5):

[4] Gusnina M, Wiharto, Salamah U. Student Performance Prediction in Sebelas Maret University Based on the Random Forest Algorithm[J]. Ingénierie des Systèmes d'Information, 2022, 27(3).

[5] SHAHIRI A M, HUSAIN W, RASHID N A. A review on predicting student's performance using data mining techniques[J]. Procedia Computer Science, 2015, 72: 414.

[6] BINMAT U, BUNYAMIN N, Arsad P M, *et al.* An overview of using academic analytics to predict and improve students' achievement; a proposed proactive intelligent intervention[C]// IEEE 5th Conference on Engineering Education, Selangor: IEEE, 2013, 126.

[7] Lan Jiafeng. Early warning of freshman academic performance based on card data[D]. Wuhan: Huazhong Normal University, 2022.

[8] Sun Meijuan, Zhang Xian, Nian Mei. Research on student profiling based on campus card and academic performance data[J]. Computer Era, 2023, (8): 20.

[9] Liu Xiaoyun, Liu Jingling, Li Zhonghao. Student performance prediction research based on multiple linear regression[J]. Development of Science and Technology in China, 2022, 32(3): 203.

[10] Lin Tingting. Research on student performance prediction model based on BP neural network[J]. Computer Technology and Automation, 2022, 41(1): 79.

[11] NASER S A, ZAQOUT I, GHOSH M, *et al.* Predicting student performance using artificial neural network: in the faculty of engineering and information technology[J]. International Journal of Hybrid Information Technology, 2015, 8(2): 221.